

A formal knowledge base for amplicon sequencing data and geochemistry of the Rub' al Khali desert

Rund Tawfiq¹, Marwa Abdelhakim¹, Sulaiman M. Alajel²,
Mohammed Alarawi¹, Hind Aldakhil³, Abderahmane Derouiche¹,
Daniela I. Drautz-Moses⁴, Michel Dumontier⁵, Raik Grünberg⁶,
Alejandra Lopez-Velazquez¹, Susana Martinez Arbas⁷, Kexin Niu¹,
Krishnakumar Sivakumar¹, Tiannyu Wang⁸, Xiang Zhao⁴,
Jood Zubair¹, Magnus Rueping⁸, Robert Hoehndorf^{1*}

¹Bio-Ontology Research Group (BORG), Computer, Electrical and
Mathematical Sciences and Engineering (CEMSE) Division, King
Abdullah University of Science and Technology (KAUST), Thuwal,
Saudi Arabia.

²Ministry of Environment, Water and Agriculture, Saudi Arabia.

³National Livestock & Fisheries Development Program (NLFDP),
Ministry of Environment, Water and Agriculture (MEWA), Riyadh,
13413, Saudi Arabia.

⁴Bioscience Core Lab, King Abdullah University of Science and
Technology, Thuwal, Saudi Arabia.

⁵Institute of Data Science, Department of Advanced Computing
Sciences, Faculty of Science and Engineering, Maastricht University,
Maastricht, The Netherlands.

⁶Biomedical Sciences Division (BioMed), King Abdullah University of
Science and Technology (KAUST), Thuwal, Saudi Arabia.

⁷NIUM, Esch-sur-Alzette, Luxembourg.

⁸Physical Science and Engineering (PSE) Division, King Abdullah
University of Science and Technology (KAUST), Thuwal, Saudi Arabia.

*Corresponding author(s). E-mail(s): robert.hoehndorf@kaust.edu.sa;

Abstract

We collected soil and plant material during five campaigns across a 1,043 km transect of the Saudi Rub' al-Khali (Empty Quarter). The resource links 2,516 non-control specimen entries with laboratory processing and a canonical amplicon table of 1,271 profiles, of which 1,237 pass control and read-depth exclusions. Associated observations comprise 274 field environmental logs, 712 archived-soil pH measurements, 71 field X-ray fluorescence sessions, 725 laboratory X-ray fluorescence records and 3,234 site-month climate summaries. We represent sites, specimens, processes and measurement values in a modular RDF knowledge base using the SemanticScience Integrated Ontology and validated chemical, environmental and taxonomic references. Shared identifiers preserve specimen identity and assay-specific provenance. Validation checks source reconciliation, measurement admission, identifier mappings, structural constraints and query outputs. The graph supports retrieval across these linked observations through a web portal and SPARQL endpoint. Versioned observations, graph modules and workflow code support comparisons across sites, campaigns and assay types.

Keywords: Knowledge graph, Amplicon sequencing, Rub' al Khali, SIO, Semantic web

1 Background and summary

The Rub' al-Khali (Empty Quarter) spans approximately 500,000 km² of the southern Arabian Peninsula¹. Its soils experience large temperature fluctuations and irregular rainfall², yet support diverse microbial communities³. Repeated observations across soil compartments and seasons provide a basis for studying these communities together with their environmental conditions.

Previous Saudi Arabian surveys examined elevation gradients⁴, rainfall responses across soil compartments⁵, and bulk and rhizosphere soils across regions⁶. Studies of desert plants further connected microbial profiles with host-associated habitats⁷. A repeated transect adds a specific data-management requirement: each community profile must remain connected to its field specimen, collection visit, laboratory processing and associated measurements. Site identity alone is insufficient for temporal comparisons because a site can have several visits and assays can use different field replicates.

Multi-campaign resources such as Tara Oceans link campaign, station and sampling-event registries to environmental and sequence data⁸. MIxS/MIMARKS specify contextual fields for environmental sequence data⁹, and STREAMS extends reporting guidance for environmental microbiomes¹⁰. The National Microbiome Data Collaborative uses persistent identifiers and shared schemas to connect environmental microbiome data¹¹. Knowledge graphs such as KG-Microbe¹² and MetagenomicKG¹³ organize taxonomic, functional and environmental information across sources. We apply these principles to the specimen and process provenance of a single landscape survey.

We collected soil and plant material during five campaigns (2023–2025) along a 1,043 km transect across the Saudi Rub’ al-Khali. The catalogue contains 60 primary sites and 10 named locations. A 2,550-row specimen catalogue contains 2,516 non-control specimen rows. The canonical amplicon table contains 1,271 profiles; the companion ecology analysis retains 1,237 after control and read-depth exclusions. The resource links these profiles to field environmental observations, archived-soil pH, field and laboratory XRF measurements, and climate data. The coverage tables distinguish specimens, profiles and assay observations.

We encoded sites, specimens, processes, qualities and measurement values with the SemanticScience Integrated Ontology (SIO)¹⁴. ChEBI¹⁵, ENVO¹⁶ and NCBI Taxonomy¹⁷ provide chemical cross-references, environmental classes and validated taxonomic identifiers. Classifier lineages retain their provenance alongside validated identifiers and explicit unresolved mappings. The resulting RDF modules support queries across these linked observations.

Technical validation examines source reconciliation, measurement admission, identifier mappings, structural constraints and executable query examples. We report the scope of each check separately, including the constructs and modules processed by ELK. The portal at <https://rubalkhali.science/> exposes the graph through site pages and a SPARQL endpoint. Data Records and Data Availability identify the distributed artifacts and outstanding release dependencies. The companion ecology manuscript reports the community, environmental-association and functional analyses¹⁸; this Data Descriptor documents their source data, representation and reuse conditions.

2 Methods

2.1 Study area and sampling design

We conducted five sampling campaigns across the Rub’ al-Khali along a transect from the western escarpment (19.15°N, 45.17°E) to the eastern border of Saudi Arabia with Oman (20.71°N, 54.73°E). The 60 ordered sampling sites yield a 1,043-km cumulative path and lie 1,015 km apart end to end. Figure 1 shows the site coordinates and altitude profile. Cumulative distance sums great-circle distances between successive sites, using an Earth radius of 6,371 km.

The transect comprises 60 primary sampling sites at predetermined coordinates and 10 named locations, including wells, a hot spring and an oil-spill site. Trip 1 source identifiers 61–64 are aliases for four named catalogue locations, established by coordinate matching. The repeated-campaign analytical frame uses sites 1–60. The metadata link each observation to its site, campaign and specimen.

To capture temporal variability, we conducted campaigns across different seasons: late winter 2023 (Trip 1, 64 sites visited), summer 2023 (Trip 2, truncated to 8 sites due to extreme temperatures), winter 2024 (Trip 3, 65 sites visited), summer 2024 (Trip 4, 60 sites visited), and autumn 2025 (Trip 5, 62 sites visited). Field observations provide temperature, atmospheric pressure and relative humidity with campaign-specific availability. We summarize the observational coverage in Table 1. We acquired hand-held field XRF measurements during Trip 5 and laboratory XRF measurements of archived material from all five campaigns. The curated environmental table retains

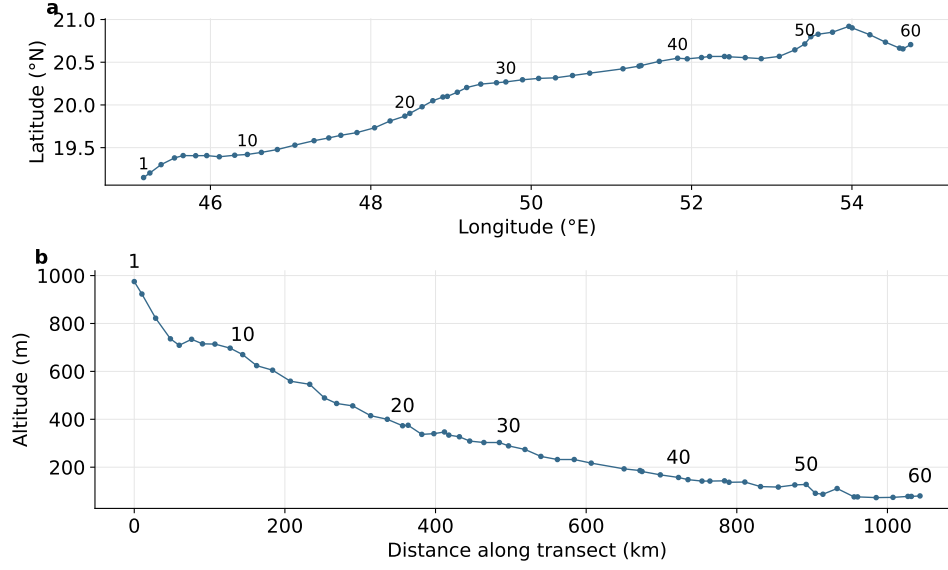


Fig. 1 Primary sampling transect across the Rub' al-Khali. Panel a shows the coordinates of sites 1–60; panel b shows their archived altitude values against cumulative great-circle distance in site order. Labels identify selected sites. Coordinates and altitude come from the versioned site-altitude table; the elevation source and vertical datum remain undocumented.

274 nonblank source rows. Of these, 260 resolve to catalogue site visits and generate 260 temperature, 187 pressure and 251 humidity values. Fourteen auxiliary or revisit labels remain table-only pending unambiguous site identification. Trip 2 includes eight temperature measurements spanning 34.5–41.9°C; pressure and humidity were unavailable for that campaign. Field observations entered the dataset only when their site identity and measurement range could be established.

Table 1 Coverage of field environmental logs. Numeric identifiers, named locations and auxiliary observations retain separate source labels.

Campaign	Season	Field-log records	Measurements available
Trip 1	Late Winter 2023	1–60; 4 named; 15 auxiliary*	Temp, Press, Hum.
Trip 2	Summer 2023	1–8	Temp.
Trip 3	Winter 2024	1–60; 5 named	Temp, Press, Hum.
Trip 4	Summer 2024	1–60	Temp, Hum.
Trip 5	Autumn 2025	1–60; 2 named	Temp, Press, Hum.

*The 15 observations dated March 2023 represent Trip 1 auxiliary/revisit observations.

The sampling design targeted three soil compartments: bulk surface soil, bulk subsurface soil (nominal 5–10 cm depth, hereafter “deep”), and subsurface soil collected near roots (5–10 cm depth, hereafter “root-adjacent”). The ontology class Deep Soil Sample denotes the 5–10 cm bulk compartment, called shallow-subsurface in the companion ecology analysis¹⁸. We collected three co-located field replicates per compartment using sterile tubes and sterile plastic spoons. From Trip 2 onwards, three adjacent tubes were taped or chained together for collection, so adjacent sampling positions were separated by one external tube diameter. Trip 5 used larger tubes. Exact tube diameters, surface sampling depth, individual bulk subsurface depths and distance from roots were unrecorded. These co-located replicates share a site and compartment. We preserved samples at -20°C in the field using an Engel MR040F portable freezer. At the end of each campaign, we transferred samples to an -80°C freezer in the laboratory.

The specimen catalogue additionally contains 248 plant entries linked by campaign, site and source identifier. Tissue type, host-identification procedure, vouchers and plant-specific storage details are incomplete in the archived protocol. These entries preserve plant provenance; root-adjacent soil retains its own specimen identity.

Table 2 Source-specimen coverage by campaign and soil compartment, linked through the specimen catalogue. All catalogue sites are included; plant and control rows are excluded. Columns count source rows eligible for specimen representation, expected by the extract generator, linked to at least one run, present in the ASV table and retained in the ecological analysis. Counts apply to source rows with a profile link; sequencing-profile totals are reported separately, and repeated profiles can share a specimen row.

Trip	Compartment	Specimens	Extract-linked	Run-linked	ASV table	Ecology
1	Subsurface	192	184	111	110	108
1	Root-adjacent	192	170	102	106	104
1	Surface	192	185	108	114	113
2	Subsurface	24	23	9	8	7
2	Root-adjacent	24	18	13	13	11
2	Surface	24	22	7	6	6
3	Subsurface	180	180	141	141	141
3	Root-adjacent	180	174	154	154	154
3	Surface	180	180	156	155	155
4	Subsurface	180	63	59	59	59
4	Root-adjacent	180	60	58	58	58
4	Surface	180	62	60	60	60
5	Subsurface	180	93	57	55	55
5	Root-adjacent	180	151	133	133	133
5	Surface	180	82	30	29	29

2.2 Laboratory controls

Control coverage differed by campaign and processing stage (Table 3). Extraction blanks underwent the same procedure as biological specimens. Trips 4–5 used one blank per trip-specific extraction-day batch. Trips 1–3 used one blank per extraction kit, and Trips 1–2 included shared extraction days. The PowerSoil Pro control has a sequencing record (**Extraction-Ctrl-Pro-Trip1**, library M-25-0929); the PowerSoil blank is present in laboratory records, with no matching sequencing record located in the available records.

Table 3 Control coverage relevant to the released amplicon data. Detailed material and sequence occurrences are retained in the control registry.

Campaign	Extraction controls	PCR-stage negatives	Sequenced standards
1	Per-kit; Pro blank sequenced	Nuclease-free water	Identity unresolved
2	Shared per-kit coverage	NTC-2	Two purified-DNA standards
3	Per-kit coverage	PCRCtrl, index 728	Two whole-cell standards
4	Six sequenced blanks; three unsequenced batches	PCR blank and NTC unsequenced	Amplification-positive control only
5	18 canonical blank profiles; 17 batch-linked	Six canonical profiles	NTC Field standard sequenced by shotgun

The canonical table contains 18 Trip 5 extraction-blank profiles (EB1–EB18) and six no-template PCR controls (Negative1, Negative2, Negative4–Negative7). Seventeen extraction blanks have confirmed batch links to 217 biological profiles. The six Trip 4 extraction blanks (SEB, EB1–EB5) were sequenced on a shared run and denoised separately with the same pipeline on 30 August 2026. Three of the nine Trip 4 extraction batches had unsequenced blanks, leaving complete or partial coverage gaps at 28 sites. Batch assignments identify individual specimens; the profile inventory excludes unsequenced S34Dr1 and S40PRr1. S40PRr1 failed three amplification attempts. S34Dr1 yielded DNA but was probably missed during pooling.

No-template controls contained PCR reagents and molecular-grade water in place of template DNA. Reagent-only PCR blanks contained neither added water nor template. One reagent-only blank was prepared per amplification plate. Most showed low or undetectable concentration and adapter-dimer-dominated TapeStation profiles and were excluded from sequencing. Trip 5 reagent-only blanks were lost during handling, attributed to evaporation. Trip 4’s PCR blank and NTC were excluded because their index pairs conflicted with biological libraries on the shared run. The Trips 1–3 control libraries were sequenced on NovaSeq 6000 A01018 run 321, flowcell HVJY5DRX3, and denoised with biological libraries in July 2025; they are distributed separately

from the canonical feature table. Re-denoising on 30 August 2026 supplied the control comparison in Technical Validation.

Positive standards were reserved for genus-recovery evaluation. Trips 1–2 used the ZymoBIOMICS HMW DNA Standard D6322, introduced at PCR; Trip 3 used the whole-cell D6300 Microbial Community Standard, which also undergoes extraction. Trip 4 had a laboratory amplification-positive control assessed by gel electrophoresis and TapeStation, and Trip 5’s field D6300 standard was sequenced by shotgun metagenomics. Four sequenced standards from Trips 2–3 support the amplicon genus-recovery analysis. The Trip 1 FPosCtrl1 identity remains unresolved because its index and well disagree, so it remains excluded.

D6322 contains seven bacterial genomes at 14% genomic DNA each and one yeast at 2%; D6300 contains eight bacterial genomes at 12% each and two yeasts at 2% each. These are manufacturer genomic-DNA mass fractions. The recovery analysis compares expected and observed bacterial genera, including unassigned reads and additional taxa. Such departures can reflect processing contamination, amplification, classification or index-assignment errors. The registry retains intended materials, observed profiles and identity evidence separately.

2.3 DNA extraction and sequencing

Genomic DNA was extracted from 0.9–1.2 g of soil using either the QIAGEN PowerLyzer PowerSoil Kit (catalogue no. 12855-50) or the QIAGEN DNeasy PowerSoil Pro Kit (catalogue no. 47014). Both kits were used in Trips 1–3: among the sequenced profiles, the specimen metadata identify the PowerLyzer PowerSoil Kit for 77 Trip 1, 10 Trip 2 and 359 Trip 3 profiles (54 of the Trip 3 records name the MoBio PowerSoil kit) and the DNeasy PowerSoil Pro Kit for 217 Trip 1, 11 Trip 2 and 87 Trip 3 profiles; 46 Trips 1–3 profiles have no recorded kit. The QIAcube Connect was introduced for routine extraction starting with Trip 4; samples from Trips 1–3 were also processed using the QIAcube Connect when re-extraction was required. The extraction method and re-extraction history were recorded for each specimen when available.

Frozen soil samples were thawed on ice and processed immediately after thawing. Before homogenization, samples were incubated at 65°C for 10 min with agitation at 1,200 rpm, then homogenized using a PowerLyzer 24 Homogenizer at 4,000 rpm for one 45-second cycle. The remaining purification steps were performed either manually or using a QIAcube Connect instrument, following the manufacturer’s protocol. DNA was eluted in 50 µL of the elution buffer supplied with the respective kit.

Extraction controls followed the campaign-specific scheme above.

Extracted DNA was quantified using the Qubit dsDNA High Sensitivity assay, with fluorescence measured using a Varioskan LUX multimode plate reader (Thermo Fisher Scientific). We recorded extract concentrations and processing information for each specimen. Progression to indexing and sequencing required an amplicon of the expected size after first-round PCR. Extracted DNA was stored at –20°C until PCR amplification.

The V3–V4 region of the bacterial 16S rRNA gene was amplified using the Bakt.341F (CCTACGGGNGGCWGCAG) and Bakt.785R (GACTACHVGGGTATCTAATCC) primers¹⁹, both carrying Illumina overhang adapters. First-round PCR was performed

using TaKaRa Ex Taq DNA Polymerase (catalogue no. RR001B). Each reaction had a total volume of 50 μ L and contained 5 μ L of 10 $\times$ buffer with Mg^{2+} , 4 μ L of dNTP mix (2.5 mM of each dNTP), 2 μ L of each 10 μ M primer stock, 0.25 μ L of polymerase (5 U/ μ L), 5 μ L of extracted DNA and 31.75 μ L of nuclease-free water. The final primer concentration was 0.4 μ M per primer and 0.2 mM per dNTP, with 1.25 U of polymerase per reaction. The PCR programme was: initial denaturation at 95°C for 3 min; 35 cycles of 95°C for 10 s, 65°C for 30 s and 72°C for 1 min; and a final extension at 75°C for 5 min.

First-round PCR products were assessed by electrophoresis on a 1% agarose gel. Samples showing an amplicon at the expected size of approximately 500 bp were classified as passing and were then indexed. Samples without a visible product were classified as failed; these were re-amplified with 8 μ L and then 12 μ L of DNA, with water reduced to 28.75 μ L and 24.75 μ L, respectively. DNA input was set by volume because most extracts had low or undetectable concentrations. Samples without a product after three attempts were classified as failed; the laboratory protocol also included re-extraction at the end of the workflow.

Dual indices and Illumina sequencing adapters were added in a second-round PCR using the QIAGEN Multiplex PCR Kit (catalogue no. 206143), following the Illumina 16S Metagenomic Sequencing Library Preparation protocol²⁰: initial activation at 94°C for 15 min; 8 cycles of 94°C for 30 s, 55°C for 30 s and 72°C for 30 s; final extension at 72°C for 5 min.

Indexed libraries were purified using Agencourt AMPure XP beads at a 0.8:1 bead-to-sample volume ratio, quantified using the Qubit dsDNA High Sensitivity assay, and assessed for fragment size using an Agilent 4200 TapeStation with the D5000 ScreenTape assay. Libraries were pooled by molarity calculated from concentration and fragment size in the Core Lab pooling calculation. Supplementary Section S2 specifies the pooling calculation and the remaining run-specific metadata gaps. Amplicon pools were sequenced on an Illumina NovaSeq 6000 using an SP flow cell with a 500-cycle reagent kit to generate 2×250 bp paired-end reads. Sample processing and library preparation were performed in the PI's laboratory and sequencing was conducted at the Bioscience Core Lab at KAUST.

2.4 Sequencing data processing

We processed raw amplicon reads with nf-core/ampliseq (v2.14.0)^{21,22}, applying the DADA2 algorithm²³ for error correction and denoising, and assigning taxonomy against the SILVA database (v138.2)²⁴. The submission records contain 1,268 run rows over 1,238 sample identifiers, including 30 repeated identifiers. The feature-table export separately contains 29 identifiers with nonidentical paired columns; 28 retain both profiles after depth filtering. The data model distinguishes submitted runs, physical libraries and analytical profiles; the processing relationship of the paired export columns remains unresolved. This resulted in a canonical feature table of 351,472 amplicon sequence variants (ASVs) prior to contaminant removal, covering samples from Trips 1–5 and 24 control samples. Of its 1,271 profiles, 1,247 are biological; quality filtering excludes the 24 control profiles and 10 biological profiles with fewer than 1,000 reads, leaving the 1,237 profiles used by the companion ecology analysis. The

complete table is distributed unfiltered. The sequencing inventory links all 1,271 canonical profiles to paired-read inputs: 1,013 from the July processing group and 258 from Trip 5. A further 33 July inputs fall outside the canonical feature table. The July execution used pipeline groups A and B. Processing specified the Bakt_341F/Bakt_785R primers, the NovaSeq option, DADA2 SILVA v138.2 assignment and skipped QIIME steps. The container digest, full denoising parameters, physical run allocation and profile-merging provenance remain incomplete for exact regeneration from raw reads.

We compared ASV prevalence in the 17 eligible Trip 5 extraction blanks with prevalence in the 217 batch-linked biological profiles²⁵. Presence means a read count greater than zero. For each of the 351,472 ASVs, the one-sided Fisher test compares present/absent profile counts in these two groups. ASVs absent from all eligible blanks receive a score of one. The exploratory candidate rule requires $p < 0.1$ (uncorrected), presence in at least two blanks, and higher prevalence in blanks than in biological profiles. It identifies 351 candidates and applies their removal only to the 217 linked biological profiles. Prevalence is estimated across batches, each represented by one blank; the canonical table remains unfiltered.

Threshold sensitivity analysis also evaluated $p < 0.01$ and $p < 0.05$, yielding 221 and 300 candidates with the two-blank rule. At $p < 0.1$, requiring one or three blanks gives 12,051 or 175 candidates. This sensitivity documents the role of the minimum-blank requirement. The control-recovery analysis separately uses four standards from Trips 2–3.

The Trip 4 screen applies the same rule independently to six mapped blanks and 95 biological profiles: one-sided Fisher $p < 0.1$, presence in at least two blanks and greater prevalence in blanks. It identifies seven candidates, whose removal is restricted to these 95 profiles. Together, the Trip 4 and Trip 5 screens cover 312 biological profiles.

We inferred gene and pathway profiles from marker genes with PICRUST2 v2.4.1²⁶. Two executions used the Trips 1–4 ampliseq output and Trip 5 output (330,830 ASVs). KO hidden-state prediction used the PIC method, edge exponent 0.5 and seed 100. EC/KO metagenome prediction required 100 reads; pathway inference used the EC predictions. The command sequence and output identities are included in the functional-prediction provenance described in Section 3. The Trips 1–4 input sequences/table and their checksums remain unavailable. The merged products contain 1,270 profile columns, whose mapping is distributed with the functional tables.

2.5 Taxonomy normalization and mapping

A taxonomic label can denote different taxa across databases and releases, and a taxon can have several labels²⁷. We therefore retain the original SILVA lineage alongside a validated NCBI Taxonomy identifier or a lineage-scoped project identifier. Taxonomic assignments come from SILVA v138.2 (Section 2), and we treat the original classifier lineage and ASV identifier as the primary evidence. These assignments are seven rank fields (domain–species), and some are followed by a separate species assignment field and a numeric confidence. We assert the species from the ranked lineage and retain the additional species call in a provenance annotation. We normalize the seven ordered rank fields by removing rank prefixes and representing empty and unclassified entries explicitly.

The mapping procedure evaluates the SILVA-to-NCBI candidates against a pinned NCBI Taxonomy release (the OBO Foundry NCBITaxon release 2025-03-13, built from the NCBI organismal taxonomy of 2025-03-01). An identifier is retained only when it resolves, has a compatible rank and label or documented synonym, and its ancestor chain is compatible with all retained higher-rank identifiers in that source lineage. A missing, ambiguous or incompatible candidate is replaced by a deterministic project class scoped to the complete truncated source lineage and rank. Each decision records the source lineage and rank, candidate identifier, validation outcome, final identifier and reason. The output mapping covers every lineage represented in both the taxonomy export and the feature table; otherwise generation stops.

Project identifiers are scoped to one normalized rank, label and parent context. The mapping builder assigns contextual classes wherever those contexts differ. It produces the validated mappings, their decision provenance, a project taxonomy module and input/output checksums. The abundance generator accepts only mappings that pass these checks.

Queries retain the original SILVA lineage and independently validated NCBI or project identifiers. The mapping provenance reports rejected candidates and their reasons. Algorithmic checks establish consistency with the specified name, rank and ancestry rules; semantic mapping accuracy requires a separately annotated validation sample.

2.6 Shotgun-derived data

We distribute genome-relative-abundance profiles for 150 shotgun libraries and eggNOG-mapper annotations of the genome catalogue for comparison with amplicon-derived functional profiles in the ecology analysis. CoverM²⁸ reports each genome's abundance as a percentage of reads, retaining the unmapped fraction separately. The catalogue contains 991 genomes. We annotated the concatenated predicted proteome with eggNOG-mapper v2.1.12²⁹ on the KAUST Ibex cluster on 14 June 2026. The intersection of annotation and abundance data yields a 990-genome by KEGG Orthology copy-number matrix.

The ecology analysis compares whole-profile KO abundances in 125 libraries with matched amplicon profiles. Its marker-gene comparison uses 975 genomes and 119 matched libraries under the marker-specific annotation and abundance requirements. These data support downstream re-execution. Reproduction from raw shotgun reads additionally requires sequence accessions, genome and proteome catalogues, and the assembly, binning, dereplication and profiling workflows. These upstream inputs and parameters remain unavailable in the distributed resource.

2.7 PMA-treated aliquots

The PMA dataset contains nine treated/untreated pairs from three site-compartment groups at two Trip 5 campsites: C1 root-adjacent, C2 root-adjacent and C2 surface soil. For each group, three soil subsamples from one parent collection tube were prepared as separate slurries. Each slurry was divided into treated and untreated aliquots, giving three preparation pairs per parent tube. Suspensions used PBS at pH 7.4 and a 1:10

mass-to-volume ratio. Treated aliquots received 50 μM PMAxx. Both treated and untreated aliquots incubated for 10 min in darkness and were illuminated at 450 nm for 35 min at 15–20 cm, with rotation at 15-minute intervals. Samples were frozen at -20°C ; DNA was extracted from 860 μL using the PowerSoil Pro kit.

The paired marker-gene profiles describe treatment responses within these three parent specimens under matched dark and light exposure.

2.8 Environmental and climate data

We retrieved historical climate data from the Open-Meteo archive endpoint³⁰ for each sampling site using its geographic coordinates, yielding 12,936 monthly values: four variables over 49 months at 66 resolved sites, or 3,234 site-month rows. The variables are temperature, total precipitation, rainfall and relative humidity; annual summaries are distributed separately. We ran two separately configured acquisitions: the monthly product covers 1 January 2022 to 20 January 2026 and includes mean relative humidity, whereas the daily product covers 1 January 2022 to 1 February 2026 and reports temperature, total precipitation and rainfall. January 2026 is a partial month in the monthly acquisition. The acquisition metadata identify the query configuration and the retained climate values. Original response bodies, retrieval timestamps and the model/product version remain unavailable for the monthly acquisition. Reproduction uses the distributed climate observations; a new API request constitutes a new acquisition.

Daily rainfall exposures in the ecology analysis use the sampling-site coordinates and the distributed Open-Meteo time series. A five-product comparison combines Open-Meteo with CHIRPS, CMORPH, IMERG and NASA POWER; its provenance identifies the grid cells and 21 source datasets used. The daily exposures and monthly summaries are distinct data products with their own acquisition metadata.

Daily NASA POWER PRECTOTCORR precipitation covers sites 1–60 from 1 January 2022 to 31 December 2025 (87,660 site-days). Its original API request metadata were not preserved. Sites 51–55 share one POWER grid cell.

Field researchers manually classified each site according to its biome (e.g., desert biome) and specific environmental features (e.g., sand dune, saline pan, gravel, desert oasis). We subsequently mapped these classifications to the Environment Ontology (ENVO)¹⁶.

2.9 XRF acquisition, processing and reconciliation

We acquired XRF data through two complementary workflows. During Trip 5, field researchers used an Olympus Vanta VMR-XXX-G3-E handheld analyser (serial 841307) for in-situ measurements. The source log contains 106 entries across 59 sites; 71 complete sessions at 58 sites were retained, and nine of those sites have repeated complete sessions. The field log identifies measurement sessions and sites. Field and laboratory observations are paired by site. Archived material from all five campaigns was subsequently measured in the laboratory. The laboratory inputs contain 547 selected sample records from Trips 1–4 and 178 from Trip 5 (725 records in total), and all 725 are used in the analyses.

Laboratory measurements used a Vanta handheld analyser in Fast Mode¹⁸. All 178 Trip 5 specimens used Fast Screening-He8mm with helium and an 8 mm diameter. Element and oxide-equivalent channels are retained separately. The analytical dataset uses the Fast Screening observations for V4Dr3 and V18Dr3. The exact gas and diameter settings for Trips 1–4 remain undocumented. Suppl. Section S3 describes acquisition and observation selection.

The laboratory acquisition metadata lack the exact model variant, serial number, measurement duration, tube settings, sample preparation, concentration units and calibration results. Statistical-error percentages and detection-limit units are distinct from concentration units. We retain instrument-reported magnitudes with undocumented concentration units. At Site 4, field readings targeted soil and a second substrate described as salt or oil; its substrate identity remains unresolved.

Repeated analyte entries are combined using the maximum positive value for Trips 1–4 and the last reported value for Trip 5. Repeated Trips 1–4 entries have identical values. Trip 5 contains repeated Al, Na and Si measurements for V14Dr1, for which the last reported value is retained. These rules define the 725-record laboratory dataset and are evaluated against the acquisition data in Technical Validation.

The chemical audit evaluates 93 reporting channels against ChEBI release 247 (3 December 2025) and the PubChem property snapshot of 23 July 2026^{15,31}. It retains 79 supported ChEBI and 91 PubChem cross-references; unresolved channels have explicit null identifiers. Checks cover identifier existence, labels, formula, charge and element/oxide type. The P_2O_5 channel uses an oxide-equivalent reporting basis linked to P_4O_{10} by empirical ratio. Light Elements remains an instrument-defined aggregate channel.

For a descriptive cross-workflow diagnostic, we took the median of repeated field sessions at each site and compared it with the Trip 5 Deep laboratory record at the 58 sites measured by both workflows. The diagnostic estimates site-level rank agreement. Calibration would require paired aliquots, documented units and reference materials. Technical Validation reports the observed correlations.

2.10 Archived-soil pH acquisition and admission

The archived-soil pH dataset contains admitted measurements linked to their physical specimens and assay sessions. The ecology analysis uses the same measurement values. We measured archived dry soil after sieving below 2 mm. Each assay used an independent 1 g scoop suspended in 2.5 mL 0.01 M $CaCl_2$. Suspensions were mixed on a HulaMixer for 30 min and stood for 30 min at room temperature. We used an accumet AB315 meter with a Fisher Scientific accumet electrode (catalogue no. 13-620-112), and recorded the reading when the meter displayed its green stability indicator. The electrode was calibrated each morning at two points (pH 7 and 10), and we recorded the electrode slope and pH-10 buffer read-back for every session. Between samples, the electrode was rinsed along its full length with Milli-Q water, immersed in a separate container of Milli-Q water and dried by gently tapping with Kimwipes. A measurement was admitted only if its session was calibrated, the electrode slope was between 95% and 102%, and the read-back was within 0.1 pH units of nominal.

Admission additionally requires a numeric pH between 0 and 14 and an accepted measurement date on or before 3 August 2026. Assay dates must resolve unambiguously to calendar dates; observations with ambiguous or post-cutoff dates are excluded. Threshold endpoints are inclusive. Accepted measurements span 13 July to 2 August 2026. Of 1,168 specimen-assay entries, 712 measurements met the admission criteria. The remainder consisted of 356 rows without a measurement, 45 marked as depleted, 36 with ambiguous dates, and 19 numeric rows with incomplete quality control records. Rows with unavailable values or incomplete admission evidence retain their audit dispositions. We reconstructed 29 sessions from trip, accepted date, electrode slope and pH-10 read-back.

All 156 admitted Trip 4 pH measurements used archived field replicate 2. Each admitted measurement is linked to its physical specimen and retains its original assay identifier. Matching with community profiles is at campaign, site and compartment level. The archived RDF protocol identifies an accumet AB315 with Probe 3 and automatic temperature compensation, and specifies mixing without fine grinding.

Table 4 Admitted archived-soil pH observations by campaign and compartment. Trip 4 measurements use field replicate 2.

Campaign	Surface	Subsurface	Root-adjacent	Total
1	52	56	49	157
2	5	6	12	23
3	54	50	52	156
4	50	52	54	156
5	30	57	133	220
Total	191	221	300	712

2.11 Knowledge representation

We represent the resource as a modular RDF knowledge base. The terminology module (TBox) defines classes and relations in OWL³²; the domain modules (ABoxes) describe sites, physical specimens, processes and data products. Shared identifiers connect these modules (Table 5). SIO supplies the upper-level classes and relations¹⁴. The terminology declares 333 classes (297 project classes and 36 referenced classes), 20 object properties and 35 datatype properties. These counts apply to declarations in the terminology module. KG v3.0.1 supplies reviewed English definitions for all 342 project terms, recorded as SKOS and IAO definition annotations. Claude Sonnet (model identifier `claude-sonnet-5-5`) drafted the definitions from term labels, axioms and dataset context; OpenAI Codex performed automated review and editing. The terminology identifies four reference ontologies as imports; observation modules are linked as dataset parts. Definition drafts, model provenance and review dispositions are distributed with the release.

2.11.1 Measurement and provenance

We separate a measured quality from the information that quantifies it. A temperature quality belongs to a site; a measurement process produces a value for that quality. The value contains the numerical reading and, where documented, its unit. Counts, dimensional measurements and measurements with undocumented units use the appropriate datatype and unit constraints in their assigned shapes.

Listing 1 illustrates this distinction for a field temperature. The process belongs to a site visit, the visit targets the site, and the measured quality is an attribute of that site. The process-to-visit path supplies the observation context. SIO's value-to-quality relation and its inverse support traversal in both directions.

Listing 1 Field temperature and its visit provenance.

```
@prefix rak: <https://rubalkhali.science/kb/RAK_> .
@prefix sio: <http://semanticscience.org/resource/SIO_> .
@prefix pato: <http://purl.obolibrary.org/obo/PATO_> .
@prefix uo: <http://purl.obolibrary.org/obo/UO_> .
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .

rak:P000001 a rak:0000006 ;
  sio:000068 rak:3000001 ;
  sio:000229 rak:4000001 .
rak:3000001 a rak:0000003 ;
  sio:000291 rak:1000001 .
rak:4000001 a rak:0000010 ;
  sio:000221 uo:0000027 ;
  rak:2000003 "20.7"^^xsd:double ;
  sio:000215 rak:5000001 .
rak:5000001 a pato:0000146 ;
  sio:000011 rak:1000001 ;
  sio:000216 rak:4000001 .
```

Sampling processes output material-sample collections. Collection membership links each physical replicate to the sampling event; the specimen's derivation edge also links it directly to its site. A protocol is a reusable information specification, and a protocol execution is a process. Extraction consumes soil and produces DNA; library preparation consumes DNA and produces an amplicon library; sequencing consumes the library and produces a digital FASTQ dataset. Quality-control processes consume FASTQ datasets and produce direction-specific metrics. Suppl. Sections S1.2–S1.7 give the corresponding formal patterns and graph examples.

Site annotations use existential restrictions to relate a site to an ENVO biome or environmental feature. Such a restriction specifies a filler of the named class. The source annotation metadata provides the recorded assignment and its provenance.

Laboratory controls are occurrence-specific materials with processing roles. A control-processing execution consumes the control, realizes its role and belongs to a documented processing group. Trips 4–5 use trip-specific extraction-day batches. Trips 1–3 use per-kit controls, and Trips 1–2 include shared extraction days. Occurrence-specific identifiers and the identity metadata distinguish reused control labels. Confirmed, provisional, unresolved and inapplicable dispositions retain their supporting evidence. The D6300 whole-cell and D6322 purified-DNA standards enter the process chain at extraction and PCR, respectively.

2.11.2 Taxonomic and chemical observations

For each profile, the taxonomy generator aggregates ASV counts at seven ranks from domain to species, retaining the rank-truncated classifier lineage and its validated NCBI or project identifier. Each workflow produces fourteen datasets: read counts and relative read abundance for each rank. The relative value divides the aggregated count by the profile’s total reads; explicit unclassified lineages retain their contribution to that denominator. A taxon-rank entry adds a value and a quality linked to shared dataset, workflow, taxon and FASTQ objects. These links preserve the profile context during retrieval. The taxon IRI is used as an individual in the abundance graph and as a class in the mapping module. OWL 2 punning gives these uses separate semantics: observation links identify the measured taxon, while class-level axioms supply the reference ancestry. The generator performs the rank aggregation explicitly.

XRF processes have several concentration-value outputs. A laboratory process takes an archived specimen as input; a field process targets a site and retains the instrument test identifier. The chemical mapping supplies supported ChEBI and PubChem cross-references for reporting channels, including oxide-equivalent channels. Unit documentation and acquisition scope are described in Section 2.9.

Archived-soil pH processes consume and target the assayed specimen, produce a pH value linked to its acidity quality and belong to a reconstructed measurement session. Field-replicate substitutions require separate physical identities and an explicit campaign, site and compartment matching level (Section 2.10).

2.11.3 Formal patterns

The following Description Logic patterns³³ specify the entity relationships used by the resource. The notation uses $\sqsubseteq$ for subclass inclusion, $\sqcap$ for conjunction, $\exists R.C$ for at least one relation R to a member of class C , and $\forall R.C$ to restrict all R fillers to C . Structural validation checks the corresponding data constraints. Suppl. Sections S1.2–S1.7 provide concrete RDF examples.

The sampling pattern links a process to its target site and output collection, whose members are physical samples.

$$\text{SamplingProcess} \sqsubseteq \text{Process} \quad (1)$$

$$\text{Sample} \sqsubseteq \text{MaterialEntity} \quad (2)$$

$$\text{SampleCollection} \sqsubseteq \text{Collection} \sqcap \forall \text{hasMember}.\text{Sample} \quad (3)$$

$$\text{SamplingProcess} \sqsubseteq \exists \text{hasTarget}.\text{Site} \sqcap \exists \text{hasOutput}.\text{SampleCollection} \quad (4)$$

$$\text{Sample} \sqsubseteq \exists \text{isDerivedFrom}.\text{Site} \quad (5)$$

A protocol execution links an information specification to the entities consumed and produced by the process.

$$\text{ProtocolExecution} \sqsubseteq \text{Process} \quad (6)$$

$$\text{Protocol} \sqsubseteq \text{InformationContentEntity} \quad (7)$$

$$\text{ProtocolExecution} \sqsubseteq \exists \text{isSpecifiedBy}.\text{Protocol} \quad (8)$$

$$\text{ProtocolExecution} \sqsubseteq \exists \text{hasInput}.\text{Entity} \sqcap \exists \text{hasOutput}.\text{Entity} \quad (9)$$

The abundance pattern connects a sequence-analysis workflow to a collection of values and to the taxonomic qualities those values quantify.

$$\text{BioinformaticWorkflow} \sqsubseteq \text{Process} \quad (10)$$

$$\text{TaxonomicDataset} \sqsubseteq \text{Collection} \quad (11)$$

$$\text{BioinformaticWorkflow} \sqsubseteq \exists \text{hasInput.SequenceDataset} \sqcap \exists \text{hasOutput.TaxonomicDataset} \quad (12)$$

$$\text{TaxonomicDataset} \sqsubseteq \forall \text{hasMember.AbundanceObservation} \quad (13)$$

$$\text{AbundanceObservation} \sqsubseteq \text{Quantity} \sqcap \exists \text{isMeasurementValueOf.AbundanceQuality} \quad (14)$$

$$\text{AbundanceQuality} \sqsubseteq \exists \text{isAttributeOf.Taxon} \quad (15)$$

$$\text{AbundanceQuality} \sqsubseteq \exists \text{hasMeasurementValue.AbundanceObservation} \quad (16)$$

XRF specializes the measurement pattern through a specimen input for laboratory measurements and a site target for field measurements.

$$\text{XRFProcess} \sqsubseteq \text{MeasurementProcess} \sqcap \exists \text{hasOutput.Concentration} \quad (17)$$

$$\text{LabXRFProcess} \sqsubseteq \text{XRFProcess} \sqcap \exists \text{hasInput.SoilSample} \quad (18)$$

$$\text{FieldXRFProcess} \sqsubseteq \text{XRFProcess} \sqcap \exists \text{hasTarget.SamplingSite} \quad (19)$$

$$\text{Concentration} \sqsubseteq \text{Quantity} \sqcap \exists \text{isMeasurementValueOf.AnalyteQuality} \quad (20)$$

The sequencing pattern composes extraction, library preparation and sequencing through their shared inputs and outputs.

$$\text{DNAExtractionProcess} \sqsubseteq \text{ProtocolExecution} \quad (21)$$

$$\text{DNAExtractionProcess} \sqsubseteq \exists \text{hasInput.SoilSample} \sqcap \exists \text{hasOutput.DNAExtract} \quad (22)$$

$$\text{LibraryPreparationProcess} \sqsubseteq \text{ProtocolExecution} \quad (23)$$

$$\text{LibraryPreparationProcess} \sqsubseteq \exists \text{hasInput.DNAExtract} \sqcap \exists \text{hasOutput.AmpliconLibrary} \quad (24)$$

$$\text{SequencingProcess} \sqsubseteq \text{ProtocolExecution} \quad (25)$$

$$\text{SequencingProcess} \sqsubseteq \exists \text{hasInput.AmpliconLibrary} \sqcap \exists \text{hasOutput.FASTQDataset} \quad (26)$$

$$\text{FASTQDataset} \sqsubseteq \text{Dataset} \quad (27)$$

$$\text{FASTQDataset} \sqsubseteq \forall \text{hasMember.SequenceRead} \quad (28)$$

Quality-control processes attach measurement values to qualities of the source sequence dataset.

$$\text{SequencingQCProcess} \sqsubseteq \text{Process} \quad (29)$$

$$\text{SequencingQCProcess} \sqsubseteq \exists \text{hasInput.FASTQDataset} \sqcap \exists \text{hasOutput.QCMeasurementValue} \quad (30)$$

$$\text{QCMeasurementValue} \sqsubseteq \text{Quantity} \sqcap \exists \text{isMeasurementValueOf.QCQuality} \quad (31)$$

$$\text{QCQuality} \sqsubseteq \exists \text{isAttributeOf.FASTQDataset} \quad (32)$$

2.11.4 OWL and RDF implementation

Groovy scripts using OWL API v5.1.20³⁴ generate the modules. The modular organization separates domain data from shared terminology³⁵. OpenLink Virtuoso³⁶ provides RDF storage and SPARQL query execution.

KG v3.0.1 serves the asserted union. A separate reasoning workflow uses the bundled rule program, v1.5.0, to compute consequences of a specified set of positive RDF rules. These rules expand equivalent classes and properties, subclass and subproperty hierarchies, instance types, domains, resource-valued ranges, inverse and symmetric properties, and transitive properties. The schema audit generates rules for each nonempty finite property-chain pattern in the loaded schema, including inverse operands. Additional rules introduce and classify value restrictions, classify existential restrictions from existing edges and typed fillers, propagate universal restrictions to existing resource fillers, and handle intersections, union membership and resource enumerations. All conclusions use existing RDF terms. The schema audit lists the loaded imports and the constructs outside this rule set.

The asserted graph contains the deduplicated union of the loaded modules. Rules read this graph together with a separate inferred graph and insert only conclusions absent from both. Each update selects a bounded batch of distinct conclusions; each rule repeats until its next batch adds zero triples. The complete schedule repeats until every rule adds zero triples in one full pass. This stopping criterion defines the least fixed point of the emitted rule program.

The controller preserves the rule text, partition inventories, derived-triple counts, runtime versions and input identifiers. Query truncation, SQL errors and exhaustion of the pass limit terminate the run with a failure status. After convergence, separate diagnostics test explicit disjoint-class and complement clashes, membership in `owl:Nothing`, and incompatible laboratory-control roles. These diagnostics and the finite rule set define the scope of operational entailment validation.

2.11.5 Structural constraints

ShEx shapes³⁷ check selected nodes for required types, properties, cardinalities and datatypes. The sampling-site shape requires a site type, label and geometry predicate. Its unconstrained geometry object checks presence; validation of the WKT datatype and coordinate syntax is a separate requirement. Additional type assertions are accepted independently of how they were obtained.

The sample shape requires a replicate type, a label, a source identifier and an IRI-valued derivation edge. The DNA concentration shape checks its numeric value, nanograms-per-microlitre unit and quality/process links. The XRF-process shape requires a protocol, one or more values and an input or target IRI. Checking the class of that input or target requires a corresponding target shape or scientific-invariant check. These open shapes admit predicates outside their declared constraints; the negative fixtures test the specific failures reported in Technical Validation.

2.12 Schema validation

We checked RDF structure using Shape Expressions (ShEx)³⁷. ShEx provides a formal structural schema language that validates the presence, cardinality, and datatype of RDF properties. We defined specific ShEx shapes for all core entity types in the knowledge graph. These entity types include sampling sites, soil samples, DNA extracts, sequencing datasets, measurements, quality control records, laboratory control materials, control roles, laboratory batches, control sequence occurrences, archived-soil pH processes, values and sessions, and taxonomy mappings.

We integrated this schema validation into our deployment workflow using a Shape Map. The Shape Map functions as a binding directive, explicitly linking node selectors (e.g., all instances of the sampling site class) to their respective ShEx shape definitions. The workflow tests each newly generated segment of instance data against the schema. It terminates the deployment if it detects missing mandatory properties (e.g., GPS coordinates for a sampling site) or invalid datatypes. Each passing test establishes conformance to the assigned shape’s specified constraints. We implemented this validation using the Apache Jena ShEx library (v4.10.0) invoked from custom Groovy scripts. The multi-gigabyte taxonomy ABox is excluded from this memory-bounded ShEx run; the exact generated file is instead parsed in full with an independent Turtle parser and checked by a streaming validator over every classification process–dataset–quality–value relationship.

2.13 Logical consistency (OWL reasoning)

We ran ELK v0.4.3³⁸ through OWL API v4.5.26, loading the tractable modules into a merged ontology. This operational check processes the constructs supported by ELK. The modules also contain universal restrictions, inverse properties and other constructs outside that support. An explicit OWL profile report and checked projection are required to establish the logical scope of the result. The run excludes the multi-gigabyte taxonomy ABox and large reference modules. We report the checked subset and the return value separately from structural validation in Technical Validation.

2.14 Reproducible workflow

The Nextflow workflow orchestrates module generation, validation and manuscript-derived statistics. Local and cluster profiles configure execution infrastructure. The versioned KG release workflow regenerates the scientific modules, checks their hashes against the frozen manifest, loads a fresh asserted database and validates source-derived query results before switching the public service. Upstream sequence processing has its own sample inventories, workflow parameters and environments.

The release manifests identify fixed source artifacts and generated modules. Reproduction has three stages: downstream analyses from curated inputs, generation of those inputs from primary records, and replication of the field/laboratory measurements. The package’s workflow evidence covers the executed stages; missing upstream sequence-processing inputs and acquisition metadata are stated in their respective Methods subsections. A complete release requires a fresh-checkout execution record,

checksum comparisons and successful main/supplement builds against the same inputs.

3 Data records

The source and derived-data package contains graph modules, curated tables, specimen and assay provenance, validation evidence and workflow code. The machine-readable manifest `PRE_RELEASE_MANIFEST.tsv` identifies payload sizes, checksums, categories and release/licence status. Separately retrieved bulk payloads are listed in the bulk manifest. Data Availability reports access routes and raw-sequence availability.

KG v3.0.1 contains 45,707,660 asserted triples from 16 manifested modules and is archived at Zenodo³⁹. The release distributes the original RDF modules, the deduplicated asserted graph in compressed N-Quads format, and a rebuild kit containing load/export code, definition provenance and validation evidence. The [release manifest](#) identifies their contents, sizes and checksums. The kit reconstructs the asserted union from the distributed modules. Scientific producer code and inputs are maintained separately in the reproducibility repository.

3.1 Knowledge-graph modules

The primary artifacts are the knowledge-graph modules, distributed as RDF/XML and Turtle files and structured by data type (Table 5). The base ontology (`rubalkhali.owl`) holds the project terminology. Domain modules share site, specimen and process identifiers; the release also includes the generated taxonomic vocabulary and four pinned reference ontologies. These modules load together into an RDF store. Domain queries require the corresponding shared terminology and linked site, specimen or process modules.

3.2 Tabular data products

The specimen catalogue links site and campaign to specimen, extraction, library, sequence run, quality-control decision and derived community. Identifiers 61–64 denote four named sites sampled only in Trip 1 and account for 36 source specimens. Assay tables provide the canonical amplicon profiles, taxonomic mappings, field environmental observations, laboratory and field XRF measurements, and archived-soil pH values.

The canonical feature table contains 1,271 profiles. The ecology analysis retains 1,237 (1,227 profiles from sites 1–60), after excluding 24 controls and ten biological profiles with fewer than 1,000 reads, including one `T1Dr1` run with 934 reads. The pH release `EQ-PH-SHARED-v1.0.1` provides 712 measurements and their specimen and session links.

Four additional products support downstream ecology analyses: the paired PMA table (`metadata/relic-dna/PMA_ASV_table.tsv`; 18 analytical columns from nine treated/untreated pairs, with controls kept separately), 150 CoverM profiles (`metadata/metagenome/coverm_profiles.tar.gz`), eggNOG annotations (`metadata/metagenome/eq.emapper.annotations.gz`) for the 990-genome analytical subset,

Table 5 The 16 RDF modules included in KG v3.0.1.

Module	Content
<code>rubalkhali.owl</code>	Base ontology (TBox)
<code>rubalkhali_sites.owl</code>	Sampling sites with coordinates and biome annotations
<code>rubalkhali_samples.owl</code>	Soil and plant specimen individuals with compartment types
<code>rubalkhali_measurements.owl</code>	Field environmental measurements and climate records
<code>rubalkhali_xrf.owl</code>	XRF measurement values from the field and laboratory workflows
<code>rubalkhali_dna.owl</code>	DNA extracts with concentration data
<code>rubalkhali_sra.owl</code>	Sequencing run records linked to ENA accessions
<code>rubalkhali_qc.owl</code>	Sequencing quality metrics (GC content, duplicate rate, sequence counts)
<code>rubalkhali_controls.ttl</code>	Laboratory-control materials, roles, batches and sequence occurrences
<code>rubalkhali_ph_eq_ph_shared_v1_0_1.ttl</code>	Archived-soil pH measurements, sessions and reconciled specimen links
<code>rubalkhali_taxonomy_abox.ttl</code>	Taxon observations over feature profiles
<code>ecosystem_module.ttl</code>	Vocabulary supporting canonical taxonomic mappings
<code>sio.owl</code>	Semanticscience Integrated Ontology v1.59
<code>envo.owl</code>	Environment Ontology, 20 October 2025
<code>pato.owl</code>	Phenotype and Trait Ontology, 14 May 2025
<code>uo.owl</code>	Units Ontology, 25 May 2023

and six source tables for the MAG-derived versus PICRUSt2 comparison (`metadata/metagenome/measured_function_inputs.tar.gz`). The daily climate inputs are selected by `metadata/climate/current_analysis_inputs.json`; the monthly acquisition is documented in `metadata/climate/climate_acquisition_frozen.json`. The complete field environmental observations are provided in `metadata/environmental/environmental_measurements_curated.tsv`. The sequence inventory and functional-prediction provenance are distributed under `metadata/amplicon/` and `metadata/functional/`. Methods identifies the upstream resources required for reproduction from raw reads.

3.3 Validation schemas, queries and evidence

The reproducibility repository includes the artifacts needed to re-validate the resource: the ShEx shapes that define the structural contract for each entity family, the positive and negative structural fixtures, the archived competency SPARQL queries, and the machine-readable evidence bundles (reconciliation summaries, audit tables, validation records and checksum manifests) produced by the release workflow. A companion data dictionary (`metadata/DATA_DICTIONARY.tsv`) records the field names, data

types, units, missing-value conventions and definitions for the curated canonical tabular records. Quantities carry units only where a unit is documented at the source: environmental source tables retain degrees Celsius, hectopascals (millibars) and percent relative humidity; in the graph, pressure magnitudes are multiplied by 100 and asserted in pascals (`obo:UO_0000110`), climate records use degrees Celsius and millimetres, and DNA concentrations are in nanograms per microlitre. XRF values are instrument-reported magnitudes with undocumented concentration units.

The manifests and data dictionary provide the versioned file-level inventory. The KG citation identifies the immutable v3.0.1 deposit (Section 6); the repository manifests additionally cover the broader analytical inputs.

4 Technical validation

Technical validation separates source reconciliation and measurement checks from structural constraints, executable queries and the operational ELK check. The evidence bundles identify the input snapshot and checks executed.

4.1 Assay validation

The PowerSoil Pro extraction-control profile contained 1,165 ASVs, with 84% of reads assigned to ASVs shared with biological profiles. Its nearest canonical profile was 46Dr1, with Bray–Curtis dissimilarity 0.39. We excluded this control from the contaminant screen. A library-order test compared nearby library numbers with different-site pairs matched by geographic-distance decile and compartment, using 999 library-order permutations. The Trip 1 and Trip 3 Bray–Curtis contrasts were -0.003 ($p = 0.15$) and $+0.006$ ($p = 0.99$), respectively. These contrasts summarize the tested preparation-order association.

For XRF, the source audit reproduces 13,390 Trips 1–4 and 4,293 Trip 5 reported cells under their stated selection rules. Repeated field sessions were reduced to site medians and paired with the Trip 5 subsurface laboratory observation at 58 sites. Among 12 analytes with positive values in both workflows at more than 10 paired sites, the median Spearman correlation was 0.21 (range -0.40 to 0.48). This diagnostic describes site-level rank agreement among the retained positive pairs.

4.2 Competency questions and SPARQL queries

Competency questions define the intended retrieval tasks (Table 6). The archived query workload covers site, geochemical, molecular-provenance and taxonomic data. The Site 10 XRF query has an archived 46-row result. Source-derived expected result tuples also specify the complete 48-observation genus profile for run `ERR16061083`, 360-row environment–taxon join, 70 site coordinates and six PCR no-template control roles. The release validator compares full result multisets, preserving duplicate rows and requiring exact identifiers. Numeric comparisons use absolute and relative tolerances of 10^{-12} ; coordinate comparisons use an absolute tolerance of 10^{-10} degrees. The query comparisons test the links between observations and their specimen, process and site context.

For KG v3.0.1, release validation verifies all 16 input-module hashes and parses the complete 45,707,660-quad export, checking every graph context. The terminology validation checks the term axioms and one definition per annotation property for each of 342 project terms. Public-release checks verify equivalent ontology content in RDF/XML, Turtle, JSON-LD and N-Triples, resolution of the four reference imports, manifest integrity, the live asserted-graph count and the deposited artifacts. Section 3 identifies the validation evidence and its input versions. Structural tests evaluate the scientific modules separately.

Table 6 Competency questions defining the retrieval workload.

ID	Competency question
CQ1	What are the XRF analyte concentrations measured at a given sampling site?
CQ2	Which sampling sites are recorded, and what are their GPS coordinates?
CQ3	Which biome and environmental features are assigned to each sampling site?
CQ4	What are the instrument-reported Light Elements values across the XRF measurements?
CQ5	Which DNA extraction records (extract, concentration, protocol) exist for a given sample?
CQ6	What is the full provenance chain from a soil sample to its sequencing run?
CQ7	What are the complete genus-level read counts and relative read abundance for run ERR16061083?

We show several example SPARQL queries. Listing 2 retrieves XRF measurements for a specific site, requiring traversal of the measurement process, quality, and value entities.

Listing 2 SPARQL query to retrieve field XRF measurements for Site 10. The query starts from the field-XRF measuring processes whose target is the site and walks to the individual analyte concentrations.

```

PREFIX rak: <https://rubalkhali.science/kb/RAK_>
PREFIX sio: <http://semanticscience.org/resource/SIO_>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

SELECT ?processLabel ?siteLabel ?analyte ?concentration
WHERE {
  ?process a rak:0000025 ;
    rdfs:label ?processLabel ;
    sio:000291 ?site ;
    sio:000229 ?value .
  FILTER(CONTAINS(LCASE(STR(?processLabel)), "field xrf"))
  ?site rdfs:label ?siteLabel .
  FILTER(?siteLabel = "Site 10")
  ?value sio:000215 ?quality ;
    rak:2000012 ?concentration .
  ?quality a ?qualityClass .
  ?qualityClass rdfs:label ?analyte .
  FILTER(?qualityClass != rak:0000032)
}
ORDER BY ?processLabel ?analyte

```

The query selects field-XRF processes targeting Site 10, then retrieves each output and its analyte quality. It returns one row per process/analyte combination, excluding the aggregate Light Elements channel. The concentration units are undocumented.

The query returned 46 analyte-value rows for Site 10: 24 from Test 5847 and 22 from Test 5848. Validation compares the returned identifiers and values with the expected observations from the site and XRF data.

Listing 3 retrieves the complete genus profile for run ERR16061083, joining across bioinformatic workflows, FASTQ datasets, and taxon observations.

Listing 3 Complete genus-level profile for run ERR16061083, ordered by descending read count and lineage.

```
PREFIX rak: <https://rubalkhali.science/kb/RAK_>
PREFIX sio: <http://semanticscience.org/resource/SIO_>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>

SELECT ?proc ?fastq ?runLabel ?lineage ?taxon ?taxonLabel
       ?count ?relativeAbundance
WHERE {
  VALUES ?runLabel { "FASTQ dataset for ERR16061083" }
  ?proc a rak:0000071 ;
        sio:000230 ?fastq ;
        sio:000229 ?dsAbs, ?dsRel .
  ?fastq rdfs:label ?runLabel .
  ?dsAbs a rak:0000074 ; rdfs:label ?absLabel ;
        sio:000059 ?valAbs .
  ?dsRel a rak:0000075 ; rdfs:label ?relLabel ;
        sio:000059 ?valRel .
  FILTER(STRENDS(STR(?absLabel), "(Genus)"))
  FILTER(STRENDS(STR(?relLabel), "(Genus)"))
  ?valAbs rak:2000026 ?count ;
        rak:2000025 ?lineage ;
        sio:000215 ?qualAbs .
  ?valRel rak:2000020 ?relativeAbundance ;
        rak:2000025 ?lineage ;
        sio:000215 ?qualRel .
  ?qualAbs sio:000011 ?taxon, ?fastq .
  ?qualRel sio:000011 ?taxon, ?fastq .
  ?taxon rdfs:label ?taxonLabel .
  FILTER(?taxon != ?fastq)
}
ORDER BY ?runLabel DESC(xsd:double(?count)) ?lineage ?proc
```

The query binds both count and relative-abundance values to the same workflow, source FASTQ dataset, taxon and classifier lineage. Dataset labels encode rank through the suffix (Genus). The query uses that emitted convention explicitly and returns all 48 observations for the named run. The source-derived expected table contains the complete profile, including each classifier lineage. The all-run query retrieves profiles from a locally loaded snapshot. That unrestricted export is a long-running operation. The environmental example below additionally binds field temperature and specimen collection to the same site visit.

Listing 4 SPARQL query to retrieve *Pseudomonas* relative abundance at sites with high temperature.

```
PREFIX rak: <https://rubalkhali.science/kb/RAK_>
PREFIX sio: <http://semanticscience.org/resource/SIO_>
```

```

PREFIX pato: <http://purl.obolibrary.org/obo/PATO_>
PREFIX ncbitaxon: <http://purl.obolibrary.org/obo/NCBITaxon_>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT ?visit ?visitLabel ?site ?siteLabel ?sample ?fastq
      ?temperatureProcess ?tempVal ?temp ?lineage ?relAbundance
WHERE {
  ?visit a rak:0000003 ; rdfs:label ?visitLabel ;
        sio:000291 ?site .
  ?site rdfs:label ?siteLabel .
  ?temperatureProcess sio:000068 ?visit ;
        sio:000229 ?tempVal .
  ?tempVal rak:2000003 ?temp ; sio:000215 ?tempQual .
  ?tempQual a pato:0000146 ; sio:000011 ?site .
  FILTER(?temp > 35.0)
  ?sampling a rak:0000024 ; sio:000068 ?visit ;
        sio:000291 ?site ; sio:000229 ?collection .
  ?collection sio:000059 ?sample .
  ?ext sio:000230 ?sample ; sio:000229 ?dna .
  ?libPrep a rak:0000065 ; sio:000230 ?dna ;
        sio:000229 ?lib .
  ?seq a rak:0000066 ; sio:000230 ?lib ;
        sio:000229 ?fastq .
  ?bioinf a rak:0000071 ; sio:000230 ?fastq ;
        sio:000229 ?dsRel .
  ?dsRel a rak:0000075 ; rdfs:label ?datasetLabel ;
        sio:000059 ?valRel .
  FILTER(STRENDS(STR(?datasetLabel), "(Genus)"))
  ?valRel rak:2000020 ?relAbundance ;
        rak:2000025 ?lineage ; sio:000215 ?qualRel .
  ?qualRel sio:000011 ncbitaxon:286, ?fastq .
}
ORDER BY ?visit ?sample ?fastq ?tempVal ?lineage

```

Each result retains the collection visit, specimen, sequence dataset, temperature observation and genus lineage. A site's visits are joined through their process provenance. This provides an environmental temperature associated with the collection visit of each returned microbial profile.

Regression fixtures execute both queries on deliberately confusable inputs. The taxonomy fixture assigns the same genus different abundances in two profiles and adds a species-rank observation; the query returns exactly the two matching genus observations. The visit fixture gives one site two visits at 40 and 20°C; the temperature-filtered query returns only the specimen collected during the 40°C visit. These tests check the implemented joins; release-level validation additionally requires expected result tables for the complete modules.

4.3 Source reconciliation and measurement validation

The archived evidence workflow produced a specimen catalogue and reconciliation summaries for specimens, extracts, libraries, runs and community profiles. The 2,516 eligible source rows reconcile exactly to 2,516 generated sample individuals; the 1,647 extraction-eligible rows reconcile to 1,647 DNA extracts; and 1,242 resolved sequence rows reconcile to 1,242 amplicon libraries and 1,242 FASTQ datasets. A site-identity

check independently matches each Trip 1 numeric identifier 61–64 to a named catalogue individual by exact coordinates and rejects coordinate or IRI disagreement, accounting for all 36 corresponding source specimens.

Environmental validation accounts for 274 field observations and checks site identity, temperature, pressure and relative humidity. Values enter the curated environmental dataset only when they satisfy the measurement range and identity criteria. The XRF provenance audit reproduced all 13,390 reported Trips 1–4 cells under the maximum-positive rule and all 4,293 reported Trip 5 cells under the last-reported rule, with no mismatched reported cells, and verified that the canonical laboratory input comprises all 725 processed records. An independent chemical-identifier validation evaluated all 93 instrument channels against the pinned ChEBI release 247 and the checksummed PubChem snapshot of 23 July 2026 (Section 2.9); all rows passed the declared rules, with explicit nulls for the aggregate LE channel and unsupported oxides. The pH validation accounts for 1,168 specimen-assay entries and reproduces the five admission dispositions in Section 2.10. It checks specimen identity for every admitted measurement, including field replicate 2 for all 156 Trip 4 measurements.

4.4 ShEx schema validation

The archived ShEx run used the shapes described in Section 2.11.5. Its tractable modules passed their assigned shapes. The pH module was validated separately over all 712 measurement processes, values and acidity qualities and all 29 sessions, and the laboratory-control module passed its shape and scientific-invariant checks. Four negative fixtures each alter one constraint: missing site coordinates, an invalid value datatype, reversed value-to-quality direction, or an XRF process with neither input nor target. The released shapes reject all four fixtures. The multi-gigabyte taxonomy ABox remains outside this memory-bounded run; it is instead parsed in full by an independent Turtle parser and scanned by a streaming structural validator, which accounted for 44,528,482 triples in 5,953,690 subject blocks with zero structural violations. This checks full-file syntax and the generator’s declared structural relationships. The separate listing validator checks printed Turtle against the modules available for that run. Taxonomy and external reference checks require the corresponding pinned payloads.

4.5 Logical consistency

ELK returned `true` for the tractable scientific-module subset loaded for validation, including the ecosystem, pH and laboratory-control modules. The checked subset excludes the taxonomy ABox and large reference modules. Its execution metadata identify the loaded axioms and import handling; class-hierarchy precomputation was omitted. The ontology uses `xsd:double` literals and constructs outside ELK’s supported fragment⁴⁰; the check therefore applies to the constructs processed by that configuration.

Validation of the separate finite-rule implementation uses synthetic examples for long hierarchies, transitive cycles, both project property-chain lengths, restriction and intersection classification, and PCR-control ancestry. The tests compare bounded,

class-partitioned execution with an unbounded reference closure and check recovery from an interrupted run. Release-specific checks compare the complete control-role inventory with its source, retrieve consequences of the measurement chains and require the inferred graph to be disjoint from the asserted graph. Completion requires the zero-addition pass specified in Section 2.11.4, followed by the explicit clash diagnostics.

Label validation checks that each labelled project entity has exactly one label. The evidence identifies the input modules, their hashes and validator versions (Section 3).

4.6 Portal access

The portal (<https://rubalkhali.science/>) retrieves data through backend SPARQL queries and renders site and specimen views. Independent validation compares query results with source-derived expected values; a user study would additionally measure task completion, errors and usability.

4.7 Identifiers and reuse metadata

Each named entity has a project-domain IRI. The manifest identifies payload files, and the data dictionary specifies curated fields, units and missing-value conventions. RDF provenance links travel with the modules and support reuse through SIO, ENVO, ChEBI, PATO, UO and validated taxonomic identifiers⁴¹.

The public endpoint provides read access to the deployed graph. The v3.0.1 Zenodo deposit fixes the released graph, module files and rebuild kit, with checksums, scoped licences and a version-specific DOI. Data Availability identifies the separate sequence-access requirement.

5 Usage notes

The portal at <https://rubalkhali.science/> provides site pages and a SPARQL editor. The endpoint <https://rubalkhali.science/sparql> accepts SPARQL 1.1 queries. For a local, low-memory example, load the base, site and XRF modules listed in Table 5 and execute the Site 10 query in Listing 2. Compare its 46 returned observations with the distributed expected identifiers and values. Taxonomic queries also require the taxonomy module and its mapping dependencies. The named-run example retrieves the complete genus profile for ERR16061083; all-run exports require a locally loaded snapshot and a long-running join. The reproduction instructions accompanying the data provide retrieval, checksum, environment and execution details (Section 3).

5.1 Choosing observations for reuse

Use specimen, collection-visit and assay identifiers when joining observations. The primary repeated-campaign frame comprises sites 1–60; the source identifiers 61–64 refer to four named catalogue locations sampled in Trip 1. Several feature profiles share a source specimen identifier. Analyses should group these profiles by their documented specimen and processing relationships.

The control registry distinguishes assay, processing stage, extraction group and identity evidence. The Trip 5 candidate screen uses 17 extraction blanks linked to 217

biological profiles and identifies 351 ASVs. The canonical feature table retains these ASVs; the companion sensitivity analysis applies their removal to the linked profiles. Trip 4 has six separately denoised extraction blanks and three batches with unsequenced blanks, leaving complete or partial coverage gaps at 28 sites. Concentration measurements exist in the source records; any concentration-based screen requires links between specimens, extractions and concentration measurements. Four Trips 2–3 standards support genus-recovery evaluation. Trips 4–5 have extraction controls but no sequenced 16S positive-standard profiles.

Archived-soil pH describes specimens measured in CaCl_2 under the admission rules of Section 2.10. Coverage varies by campaign and compartment. All Trip 4 assays used archived field replicate 2. Their supported comparison with community profiles is at campaign, site and compartment level. The versioned specimen mapping links measurements to the confirmed specimens as described in Methods.

Field and laboratory XRF observations have separate acquisition provenance and undocumented concentration units. The released values support within-workflow descriptive comparisons; cross-workflow calibration requires documented units, paired aliquots and reference materials. Site 4 field readings include different substrates, so a site-level summary combines that local heterogeneity with instrument repeatability.

The PMA table contains three site-compartment groups at two campsites. Each group has one parent collection tube, from which three separate slurries supplied treated/untreated pairs. Descriptive paired summaries retain this preparation structure and compare dye treatment under matched dark and light exposure.

5.2 Versioning and supported operations

The deposited source and checksum manifests identify the inputs used to generate each module. Preserve these files when reproducing a reported result. Rebuilding against a newer external ontology requires identifier and mapping validation, followed by regeneration of the affected modules and query results. Version-specific manifests identify the data used in each release.

KG v3.0.1 serves the asserted graph at <https://rubalkhali.science/graph/asserted/3.0.1>. This graph is the default dataset for portal and SPARQL queries. Explicit `FROM`, `FROM NAMED` and SPARQL protocol dataset parameters select the dataset for an individual request. The service evaluates stored assertions directly and rejects unavailable query modes.

The bundled finite-rule implementation supports the separate reasoning workflow described in Section 2.11.4. Publication of its output requires completion and validation of that workflow. Changes to released artifacts require a new semantic version.

For reproduction, retain the source package, module manifest, loader provenance and checksums together. The release instructions specify the pinned generation environment and coordinate-preserving import adapter. The asserted N-Quads provides the released union; the v3.0.1 rebuild kit verifies and loads the 16 distributed modules. The terminology builder uses the definitions and ontology-generation code distributed with the kit. Scientific-module generation uses the source data and code described in Section 3. Local module loading provides access to fixed inputs independently of the

portal’s deployed version. Analyses requiring additional OWL consequences need a reasoner and validation workload suited to those constructs.

6 Data availability

The project uses ENA umbrella study PRJEB104209; its submitted amplicon run records are associated with PRJEB106069. On 4 October 2026, the anonymous ENA read-run file report for PRJEB106069 returned no public run/file rows. Release of the raw reads or a complete anonymous reviewer download remains required. The source and derived-data package is maintained at <https://github.com/bio-ontology-research-group/empty-quarter-ecology-data>. Its manifests distinguish included files, separately retrieved bulk payloads and unresolved upstream provenance. KG v3.0.1, its original modules and rebuild kit are archived at <https://doi.org/10.5281/zenodo.23134168>³⁹. Project-authored RDF, ontology terms, definitions and dataset metadata in this release are licensed under CC BY 4.0. Imported vocabularies retain their respective licences, listed in the release’s third-party attribution file; project software is MIT-licensed. Raw reads and other separately licensed materials retain their own access and licensing conditions. The broader analytical tables and specimen and assay provenance are versioned in the reproducibility repository; the KG deposit contains the files enumerated in its manifest.

7 Code availability

The web application and ontology-development code are available at <https://github.com/bio-ontology-research-group/empty-quarter>. The data repository <https://github.com/bio-ontology-research-group/empty-quarter-ecology-data> contains the manuscript sources, RDF generators, metadata and Nextflow workflow. RDF generation uses Groovy scripts with OWL API v5.1.20³⁴; the ELK validation wrapper uses OWL API v4.5.26. The repository supplies execution profiles, workflow tests, a pinned Conda recipe and a hash-locked Python environment.

Funding

Susana Martinez Arbas was supported by the Fonds National de la Recherche (FNR), Luxembourg (Project Ref. 18047750). No specific funding was received for this project.

Competing interests

The authors declare no competing interests.

References

- [1] Mandaville, J.P.: Plant life in the rub’ al-khali (the empty quarter), south-central arabia. *Proceedings of the Royal Society of Edinburgh. Section B. Biological Sciences* **89**, 147–157 (1986) <https://doi.org/10.1017/s0269727000008988>

- [2] Nelli, N., Francis, D., Sow, M., Fonseca, R., Alkatheeri, A., Bosc, E., Bergametti, G.: The wind-blown sand experiment in the empty quarter desert: Roughness length and saltation characteristics. *Earth and Space Science* **11**(6), 2024–003512 (2024) <https://doi.org/10.1029/2024EA003512>
- [3] Makhallanyane, T.P., Valverde, A., Gunnigle, E., Frossard, A., Ramond, J.-B., Cowan, D.A.: Microbial ecology of hot desert edaphic systems. *FEMS Microbiology Reviews* **39**(2), 203–221 (2015) <https://doi.org/10.1093/femsre/fuu011>
- [4] Yasir, M., Azhar, E.I., Khan, I., Bibi, F., Baabdullah, R., Al-Zahrani, I.A., Al-Ghamdi, A.K.: Composition of soil microbiome along elevation gradients in southwestern highlands of Saudi Arabia. *BMC Microbiology* **15**(1), 65 (2015) <https://doi.org/10.1186/s12866-015-0398-4>
- [5] Aslam, S.N., Dumbrell, A.J., Sabir, J.S., Mutwakil, M.H.Z., Baeshen, M.M.N., Abo-Aba, S.E.M., Clark, D.R., Yates, S.A., Baeshen, N.A., Underwood, G.J.C., McGenity, T.J.: Soil compartment is a major determinant of the impact of simulated rainfall on desert microbiota. *Environmental Microbiology* **18**(12), 5048–5062 (2016) <https://doi.org/10.1111/1462-2920.13474>
- [6] Khan, M.A., Khan, S.T.: Microbial communities and their predictive functional profiles in the arid soil of Saudi Arabia. *SOIL* **6**, 513–521 (2020) <https://doi.org/10.5194/soil-6-513-2020>
- [7] Eida, A.A., Ziegler, M., Lafi, F.F., Michell, C.T., Voolstra, C.R., Hirt, H., Saad, M.M.: Desert plant bacteria reveal host influence and beneficial plant growth properties. *PLOS ONE* **13**(12), 0208223 (2018) <https://doi.org/10.1371/journal.pone.0208223>
- [8] Pesant, S., Not, F., Picheral, M., Kandels-Lewis, S., Le Bescot, N., Gorsky, G., Iudicone, D., Karsenti, E., Speich, S., Troublé, R., Dimier, C., Searson, S., Tara Oceans Consortium Coordinators: Open science resources for the discovery and analysis of Tara Oceans data. *Scientific Data* **2**, 150023 (2015) <https://doi.org/10.1038/sdata.2015.23>
- [9] Yilmaz, P., Kottmann, R., Field, D., Knight, R., Cole, J.R., Amaral-Zettler, L., Gilbert, J.A., *et al.*: Minimum information about a marker gene sequence (MIMARKS) and minimum information about any (x) sequence (MIxS) specifications. *Nature Biotechnology* **29**(5), 415–420 (2011) <https://doi.org/10.1038/nbt.1823>
- [10] Kelliher, J.M., Mirzayi, C., Bordenstein, S.R., Oliver, A., Kellogg, C.A., Hatcher, E.L., Berg, M., Baldrian, P., Aljumaah, M., *et al.*: STREAMS guidelines: standards for technical reporting in environmental and host-associated microbiome studies. *Nature Microbiology* **10**(12), 3059–3068 (2025) <https://doi.org/10.1038/s41564-025-02186-2>

- [11] National Microbiome Data Collaborative: National Microbiome Data Collaborative: Data Standards. <https://microbiomedata.org/data-standards/>. Accessed 23 July 2026 (2026)
- [12] Santangelo, B.E., Hegde, H., Caufield, J.H., Reese, J., Kliegr, T., Hunter, L.E., Lozupone, C.A., Mungall, C.J., Joachimiak, M.P.: KG-Microbe – building modular and scalable knowledge graphs for microbiome and microbial sciences. *GigaScience*, 077 (2026) <https://doi.org/10.1093/gigascience/giag077>
- [13] Ma, C., Liu, S., Won, S., Koslicki, D.: MetagenomicKG: a knowledge graph for metagenomic applications. *Bioinformatics* **42**(7), 421 (2026) <https://doi.org/10.1093/bioinformatics/btag421>
- [14] Dumontier, M., Baker, C.J., Baran, J., Callahan, A., Chepelev, L., Cruz-Toledo, J., Del Rio, N.R., Duck, G., Furlong, L.I., Keath, N., *et al.*: The SemanticScience Integrated Ontology (SIO) for biomedical research and knowledge discovery. *Journal of Biomedical Semantics* **5**(1), 14 (2014) <https://doi.org/10.1186/2041-1480-5-14>
- [15] Degtyarenko, K., Matos, P., Ennis, M., Hastings, J., Zbinden, M., McNaught, A., Alcántara, R., Darsow, M., Guedj, M., Ashburner, M.: ChEBI: a database and ontology for chemical entities of biological interest. *Nucleic Acids Research* **36**(suppl.1), 344–350 (2008) <https://doi.org/10.1093/nar/gkm791>
- [16] Buttigieg, P.L., Morrison, N., Smith, B., Mungall, C.J., Lewis, S.E.: The environment ontology: contextualising biological and biomedical entities. *Journal of Biomedical Semantics* **4**(1), 43 (2013)
- [17] Schoch, C.L., Ciufu, S., Domrachev, M., Hotton, C.L., Kannan, S., Khovanskaya, R., Leipe, D., Mcveigh, R., O’Neill, K., Robbertse, B., *et al.*: NCBI Taxonomy: a comprehensive update on curation, resources and tools. *Database* **2020**, 062 (2020) <https://doi.org/10.1093/database/baaa062>
- [18] Tawfiq, R., Abdelhakim, M., Alajel, S.M., Alarawi, M., Aldakhil, H., Derouiche, A., Drautz-Moses, D.I., Dumontier, M., Grünberg, R., Lopez-Velazquez, A., Martinez Arbas, S., Niu, K., Sivakumar, K., Wang, T., Zhao, X., Zubair, J., Rueping, M., Hoehndorf, R.: Landscape-scale bacterial biogeography across the Rub’ al-Khali. Companion ecology manuscript in preparation (2026)
- [19] Klindworth, A., Pruesse, E., Schweer, T., Peplies, J., Quast, C., Horn, M., Glöckner, F.O.: Evaluation of general 16s ribosomal RNA gene PCR primers for classical and next-generation sequencing-based diversity studies. *Nucleic Acids Research* **41**(1), 1 (2013) <https://doi.org/10.1093/nar/gks808>
- [20] Illumina: 16s metagenomic sequencing library preparation: Preparing 16s ribosomal rna gene amplicons for the illumina miseq system (2013). Available at https://support.illumina.com/documents/documentation/chemistry_documents

tation/16s/16s-metagenomic-library-prep-guide-15044223-b.pdf; accessed 27 August 2026

- [21] Ewels, P.A., Peltzer, A., Fillinger, S., Patel, H., Alneberg, J., Wilm, A., Garcia, M.U., Di Tommaso, P., Nahnsen, S.: The nf-core framework for community-curated bioinformatics pipelines. *Nature Biotechnology* **38**(3), 276–278 (2020) <https://doi.org/10.1038/s41587-020-0439-x>
- [22] Straub, D., Blackwell, N., Langarica-Fuentes, A., Peltzer, A., Nahnsen, S., Klein-dienst, S.: Interpretations of environmental microbial community studies are biased by the selected 16s rRNA (gene) amplicon sequencing pipeline. *Frontiers in Microbiology* **11**, 550420 (2020)
- [23] Callahan, B.J., McMurdie, P.J., Rosen, M.J., Han, A.W., Johnson, A.J.A., Holmes, S.P.: DADA2: high-resolution sample inference from Illumina amplicon data. *Nature Methods* **13**(7), 581–583 (2016) <https://doi.org/10.1038/nmeth.3869>
- [24] Quast, C., Pruesse, E., Yilmaz, P., Gerken, J., Schweer, T., Yarza, P., Peplies, J., Glöckner, F.O.: The SILVA ribosomal RNA gene database project: improved data processing and web-based tools. *Nucleic Acids Research* **41**(D1), 590–596 (2013) <https://doi.org/10.1093/nar/gks1219>
- [25] Fierer, N., Leung, P.M., Lappan, R., Eisenhofer, R., Ricci, F., Holland, S.I., *et al.*: Guidelines for preventing and reporting contamination in low-biomass microbiome studies. *Nature Microbiology* **10**(7), 1570–1580 (2025) <https://doi.org/10.1038/s41564-025-02035-2>
- [26] Douglas, G.M., Maffei, V.J., Zaneveld, J.R., Yurgel, S.N., Brown, J.R., Taylor, C.M., Huttenhower, C., Langille, M.G.I.: PICRUSt2 for prediction of metagenome functions. *Nature Biotechnology* **38**(6), 685–688 (2020) <https://doi.org/10.1038/s41587-020-0548-6>
- [27] Balvočiūtė, M., Huson, D.H.: Silva, rdp, greengenes, ncbi and ott—how do these taxonomies compare? *BMC genomics* **18**(Suppl 2), 114 (2017)
- [28] Aroney, S.T.N., Newell, R.J.P., Nissen, J.N., Camargo, A.P., Tyson, G.W., Woodcroft, B.J.: CoverM: read alignment statistics for metagenomics. *Bioinformatics* **41**(4), 147 (2025) <https://doi.org/10.1093/bioinformatics/btaf147>
- [29] Cantalapiedra, C.P., Hernández-Plaza, A., Letunic, I., Bork, P., Huerta-Cepas, J.: eggNOG-mapper v2: functional annotation, orthology assignments, and domain prediction at the metagenomic scale. *Molecular Biology and Evolution* **38**(12), 5825–5829 (2021) <https://doi.org/10.1093/molbev/msab293>
- [30] Open-Meteo: Open-Meteo: free weather API. <https://open-meteo.com/> (2024)

- [31] Kim, S., Chen, J., Cheng, T., Gindulyte, A., He, J., He, S., Li, Q., Shoemaker, B.A., Thiessen, P.A., Yu, B., *et al.*: PubChem 2023 update. *Nucleic Acids Research* **51**(D1), 1373–1380 (2023)
- [32] W3C OWL Working Group: OWL 2 Web Ontology Language Document Overview (Second Edition). W3c recommendation, W3C (December 2012). <https://www.w3.org/TR/owl2-overview/>
- [33] Baader, F., Calvanese, D., McGuinness, D.L., Nardi, D., Patel-Schneider, P.F.: *The Description Logic Handbook: Theory, Implementation, and Applications*, 2nd edn. Cambridge University Press, Cambridge (2007). <https://doi.org/10.1017/CBO9780511711787>
- [34] Horridge, M., Bechhofer, S.: The OWL API: a Java API for OWL ontologies. *Semantic Web* **2**(1), 11–21 (2011)
- [35] Cuenca Grau, B., Horrocks, I., Kazakov, Y., Sattler, U.: Modular reuse of ontologies: theory and practice. *Journal of Artificial Intelligence Research* **31**, 273–318 (2008) <https://doi.org/10.1613/jair.2375>
- [36] OpenLink Software: OpenLink Virtuoso Universal Server. <https://virtuoso.openlinksw.com/> (2024)
- [37] Prud’hommeaux, E., Labra Gayo, J.E., Solbrig, H.: Shape Expressions: an RDF validation and transformation language. In: *Proceedings of the 10th International Conference on Semantic Systems (SEM ’14)*, pp. 32–40. ACM, Leipzig, Germany (2014). <https://doi.org/10.1145/2660517.2660523>
- [38] Kazakov, Y., Krötzsch, M., Simančík, F.: The incredible ELK: From polynomial procedures to efficient reasoning with $\mathcal{EL}$ ontologies. *Journal of Automated Reasoning* **53**(1), 1–61 (2014) <https://doi.org/10.1007/s10817-013-9296-3>
- [39] Tawfiq, R., *et al.*: The microbial diversity of Rub’ al-Khali (Empty Quarter). Zenodo (2026). <https://doi.org/10.5281/zenodo.23134168> . <https://doi.org/10.5281/zenodo.23134168>
- [40] Motik, B., Cuenca Grau, B., Horrocks, I., Wu, Z., Fokoue, A., Lutz, C.: *OWL 2 Web Ontology Language Profiles (Second Edition)*. W3c recommendation, W3C (December 2012). <https://www.w3.org/TR/owl2-profiles/>
- [41] Wilkinson, M.D., Dumontier, M., Aalbersberg, I.J., Appleton, G., Axton, M., Baak, A., Blomberg, N., Boiten, J.-W., Silva Santos, L.B., Bourne, P.E., *et al.*: The FAIR Guiding Principles for scientific data management and stewardship. *Scientific Data* **3**(1), 160018 (2016)